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COMMENTARY

Crossing the null does not mean "no effect": a survey of internal medicine physicians on the interpretation of effect estimates with wide confidence intervals

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Abstract

Objectives: Results with confidence intervals (CIs) crossing the null should not be interpreted as indicating no effect. We assessed how often clinicians make this error and explored predictors of correct interpretation.

Study Design and Setting: We conducted a cross-sectional survey of internal medicine residents, fellows, and faculty at McMaster University (Canada) between January and March 2025. Eighteen clinical cases were presented, including 5 cases in which the point estimate indicated benefit or harm and the CI crossed the null, 5 cases in which the point estimate indicated benefit or harm and CIs excluded the null, and eight cases indicating little or no effect. Ten cases used the null and eight cases used a minimally important difference as the decision threshold. Respondents judged whether an effect was present. Co-primary outcomes were the proportions correctly identifying (1) a benefit when the point estimate indicated benefit, but the CI crossed the null, and (2) a harm when the point estimate indicated harm, but the CI crossed the null. We used logistic regression to explore associations between respondent characteristics and correct responses.

Results: Among 109 respondents (65.3% trainees, 34.7% faculty), most lacked graduate-level research training (83.2%) and leadership or formal training in systematic reviews (76.2%). When the point estimate showed benefit or harm but the CI crossed the null, only 13% to 18% correctly recognized an effect, compared with 81% to 98% when the CI excluded the null. Similar patterns were observed when using the minimally important difference threshold. No characteristics predicted correct interpretations.

Conclusion: Clinicians commonly misinterpret a CI crossing the null as "no effect," disregarding the point estimate and decision threshold. This pervasive error risks misleading clinical decisions and guideline recommendations, underscoring the need for education on proper interpretation of CIs.

Keywords: Confidence intervals; Statistical significance; Statistical interpretation; Evidence-based medicine; Clinical decision-making; Survey

1. Introduction

Misinterpreting statistical "non-significance" as "no effect" is an entrenched issue in clinical research and practice [1]. In 2019, a Nature editorial urged researchers to "never conclude there is "no difference" or "no association" just because a *P* value is larger than a threshold such as 0.05 or, equivalently, because a confidence interval (CI) includes zero," referring to the issue as a "pervasive problem" that "must stop" [2]. More than 2 decades earlier, Altman and Bland warned that the "absence of evidence is not evidence of absence" [3]. Yet, despite these longstanding cautions, the extent to which this misconception continues to influence clinicians' interpretation of study results remains unclear.

Systematic surveys of the literature have shown that over half of published articles interpret results as "no

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Plain Language Summary

When research studies report treatment effects, they commonly include a CI, which reflects the uncertainty associated with the result. Many clinicians assume that if a CI includes ‘‘no effect,’’ the treatment must not work. This is misguided. We surveyed internal medicine residents, fellows, and faculty physicians to see how they interpret treatment effects when CIs are wide or cross commonly used decision thresholds. We found that most clinicians incorrectly concluded there was ‘‘no effect’’ whenever the CI crossed the null, even when the best estimate of the effect suggested benefit or harm. Fewer than one in five clinicians correctly identified an effect in these situations. These findings show that clinicians frequently rely on statistical significance rather than the size and direction of the estimated effect when interpreting study results. This misunderstanding may lead to poor clinical decisions, inaccurate guideline recommendations, and misleading communication with patients. Education and clearer reporting are needed to help clinicians interpret CIs correctly and focus on what the results actually mean for decision-making.

difference’’ when a 95% CI crosses the null [4–6]. The erroneous interpretation is not merely semantic. For instance, when a systematic review reports a risk ratio (RR) of 1.22 (95% CI 0.95–1.71) for serious adverse events with a new therapy compared to placebo, readers may incorrectly conclude there is no effect because the CI includes the null. They may do so by basing their judgment on whether the CI crosses the null, rather than focusing on the position of the point estimate relative to a decision threshold. Such misinterpretation can lead to inappropriate recommendations, unnecessary exposure to harm (or withholding of beneficial treatment in an alternate scenario), and distortion of treatment effects to inform shared decision-making. Guideline panels and regulatory bodies that rely on this data may further propagate erroneous interpretations into practice and policy.

Correct interpretation of effect estimates should center on the position of the point estimate relative to a prespecified decision threshold—commonly the null (no effect) or a minimally important difference (MID) threshold that reflects patient importance [7,8]. The point estimate is the best single summary estimate of the effect, while the CI conveys uncertainty. If the point estimate lies beyond the decision threshold, it represents an effect, regardless of whether the CI crosses the null; however, the width of the CI reflects the degree of uncertainty around that effect. This aligns with guidance from the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) working group [9–12].

Despite these recommendations, it remains unclear to what extent practicing clinicians and trainees apply them when interpreting real-world systematic review treatment effect estimates. Surveys of physicians in multiple specialties reveal limited understanding of biostatistics and epidemiological concepts [13–20], but no recent studies have directly tested how clinicians interpret point estimates and CIs.

To address this gap, we conducted a cross-sectional survey of internal medicine trainees and faculty members at a large Canadian academic center. We aimed to quantify the extent to which respondents correctly interpreted a treatment effect with a CI crossing or not crossing the null (or MID, depending on the decision threshold); and whether respondent characteristics such as research training or experience with systematic reviews predicted correct interpretation. Secondarily, we aimed to evaluate how respondents interpreted a treatment effect with a point estimate which is close to the null, with a CI crossing or not crossing the null.

2. Methods

2.1. Study design and setting

We conducted a cross-sectional online survey of internal medicine residents, clinical fellows, and faculty members at McMaster University (Hamilton, Ontario, Canada) between January and March 2025. We surveyed academic internal medicine physicians because they emphasize evidence-informed decision-making and routinely incorporate evidence-based medicine rounds and teaching sessions into training and practice. Research ethics approval was obtained from the Hamilton Integrated Research Ethics Board.

2.2. Participants

Eligible participants were (1) residents in the three-year core internal medicine residency training program (postgraduate years [PGYs] 1–3); (2) fellows in subspecialty internal medicine fellowship programs (typically PGY 4–6); and (3) part-time or full-time faculty members in the Department of Medicine at McMaster University. The McMaster internal medicine residency program consists of 115 trainees (approximately 35–40 per year across PGY 1–3), and subspecialty fellowship programs consist of 125 trainees. The Department of Medicine consists of approximately 1000 faculty members. To maximize generalizability, no exclusion criteria were applied.

Participation was voluntary and informed consent was obtained from all participants prior to survey initiation.

What is new?

Key findings

- Clinicians frequently misinterpret confidence intervals (CI).

What this adds to what was known?

- "Crossing the null" is often incorrectly interpreted as "no effect".
- <20% identified a non-zero treatment effect when the CI crossed the null.

What is the implication and what should change now?

- Interpretation should focus on point estimates relative to decision thresholds.
- Training should explicitly address common errors in CI interpretation.

Consenting participants were free to decline further participation at any time by exiting the survey prematurely.

2.3. Survey development

A multidisciplinary team with expertise in systematic reviews, meta-analysis, and GRADE methodology developed the survey. They designed eighteen hypothetical clinical cases that presented treatment effects of a drug on the risk of nonfatal myocardial infarction among adults with type 2 diabetes. The complete survey with correct answers highlighted is available in Appendix A. Additional details regarding case and question design are available in Appendix B.

Cases required respondents to judge effects relative to either the null or a prespecified MID threshold. Both thresholds are the simplest and most widely used by systematic reviewers and guideline developers and recommended by Core GRADE [10,12]. Point estimates were accompanied by 95% CI and, in some cases, *P* values or forest plots. Cases varied the width of the CI and whether it crossed the decision threshold. When presented with the null effect threshold, multiple-choice answers captured no effect (when the point estimate is close to or on the null effect threshold), and benefit or harm (when on either side of the null effect). When presented with the MID as the decision threshold, options captured no important effect (when between the MID and the null), or important benefit or harm (when larger than the MID on either side). Response options also incorporated qualifiers reflecting certainty in the effect, for example, no effect (or probably or possibly no effect).

We pilot tested the survey with two clinicians without graduate-level research training, two clinicians with such training, and one non-clinician methodologist. Pilot participants completed the survey independently as respondents and provided informal written feedback on wording, clarity, and interpretability of questions. In addition, two pilot participants participated in a verbal walk-through with the lead author to confirm that questions were being interpreted as intended. We used pilot testing to assess face validity and clarity, not to evaluate knowledge or correctness of responses. Based on feedback, we made minor wording changes to improve clarity; no questions were substantively rewritten or removed. For example, we adjusted response options to accommodate plain language interpretations of low, moderate, and high certainty evidence consistent with GRADE, ensuring that responses reflected interpretation of the effect estimate rather than judgments about certainty.

2.4. Survey administration

Departmental administrative staff distributed a single survey link through residency, clinical fellowship, and faculty mailing lists and newsletters. The survey was hosted on LimeSurvey (https://surveys.mcmaster.ca). To avoid subsequent case questions biasing prior responses, participants could not return to previous pages to alter their responses.

2.5. Outcomes

The co-primary outcomes were (1) the proportion of respondents who correctly identified a benefit when the point estimate indicated benefit, but the CI crossed the null, and (2) the proportion who correctly identified a harm when the point estimate indicated harm, but the CI crossed the null. Secondary outcomes were the proportions of correct responses for all other cases. We also evaluated associations between respondent characteristics and likelihood of correct responses across cases.

2.6. Statistical analysis

We summarized participant characteristics and case responses using descriptive statistics. We used multiple logistic regressions to explore associations between professional status, graduate-level training or systematic review experience (Appendix C) and correct interpretations, reporting odds ratios and 95% CIs. Analyses used Stata/MP version 18.5 (StataCorp).

This methodological survey was exploratory and did not involve formal hypothesis-testing. To ensure any regression analysis conducted was adequately powered, we aimed for approximately 150 participants to satisfy the events-per-variable rule of thumb [21].

3. Results

One hundred nine participants completed the survey, of whom 102 (93.6%) responded in full. Of those reporting demographic characteristics, respondents included 49 (48.5%) resident physicians, 17 (16.8%) clinical fellows, and 35 (34.7%) faculty members. Most had little or no graduate-level training in health research methodology (83.2%) and no leadership experience or formal training in systematic reviews (76.2%) (Table 1).

3.1. Interpretations using the null as the decision threshold

When the point estimate indicated benefit or harm and CI did not cross the null, the vast majority recognized the effect (benefit: 96/109, 88.1% correct without forest plot; 102/104 (98.1%) correct with forest plot; harm: 87/108, 80.5% correct). Accuracy of responses was lower when the CI crossed the null despite an unchanged point estimate (benefit: 14/109, 12.8% correct without forest plot; 23/104, 22.1% correct with forest plot; harm: 19/107, 17.7% correct) (Fig 1, Appendix D).

Table 1. Characteristics of respondents

Characteristic	No. (%) of respondents
Professional status	
Resident physician (PGY, 1−3)	49 (48.5%)
Clinical fellow (PGY, 4−6)	17 (16.8%)
Part-time faculty member	3 (3.0%)
Full-time faculty member	32 (31.7%)
Not reported	8
Prior training in health research methodology	
No prior formal coursework or training in health research methodology, clinical epidemiology, or evidence-based medicine	55 (54.5%)
Completed at least one formal course or training session in health research methodology, clinical epidemiology, or evidence-based medicine, but no formal Masters or PhD degree completed	29 (28.7%)
Completed a formal Masters or PhD in health research methodology, clinical epidemiology, or evidence-based medicine	17 (16.8%)
Not reported	8
Prior experience in systematic reviews	
No prior experience or formal training related to conducting a systematic review	38 (37.6%)
Prior involvement in conducting a systematic review, but no prior leadership role	39 (38.6%)
Prior leadership role in conducting a systematic review, but no formal training completed	12 (11.9%)
Formal training completed related to conducting a systematic review	12 (11.9%)
Not reported	8

PGY, postgraduate year.

For point estimates directly on the null, most respondents correctly judged ‘‘no effect’’ regardless of whether the CI was narrow (101/104, 97.1% correct) or wide (97/104, 93.3% correct). For a point estimate close to the null, most respondents correctly judged ‘‘no effect’’ when the CI did not cross the null (91/104, 87.5%) but few did so when presented with the same point estimate and CI that excluded the null (5/104, 4.8%) (Fig 2, Appendix D).

3.2. Interpretation using the MID as the decision threshold

When the point estimate indicated an important benefit and CI did not cross the MID nor the null, 101 of 103 (98.0%) respondents answered correctly. When the same point estimate was presented with CI crossing the MID for benefit but not crossing the null or the MID for harm, 57/103 (55.3%) responded correctly. When the CI crossed the null and the MID for benefit (but not the MID for harm), 28/103 (27.2%) responded correctly; and when the CI crossed all three (null and MID on both sides), only 16/103 (15.5%) responded correctly (Fig 3, Appendix D).

When presented with a point estimate that was close to or at the null, most participants recognized ‘‘no important effect’’ when the CI crossed the MIDs on both sides (94/102, 92.2% correct). Most respondents correctly interpreted a point estimate directly on the null when the CI did not cross the MID (98/102, 96.1%) but fewer correctly interpreted a point estimate close to the null with CI neither crossing the null nor the MID on either side (82/103, 79.6%) (Fig 4, Appendix D).

3.3. Predictors of correct interpretation

Multiple logistic regression found no consistent associations between professional status, graduate-level training, or systematic review experience and correct responses across cases (Appendix E).

4. Discussion

4.1. Summary of main findings

We aimed to evaluate how clinicians interpret treatment effects from a systematic review where a point estimate represents a benefit or harm, but the CI crosses the null or MID decision threshold. Our objective was to determine whether clinicians interpret treatment effects based on the point estimate relative to a clinical decision threshold, or instead base judgments on whether the CI crosses the threshold. In our survey of internal medicine residents, subspecialty clinical fellows and faculty members, we

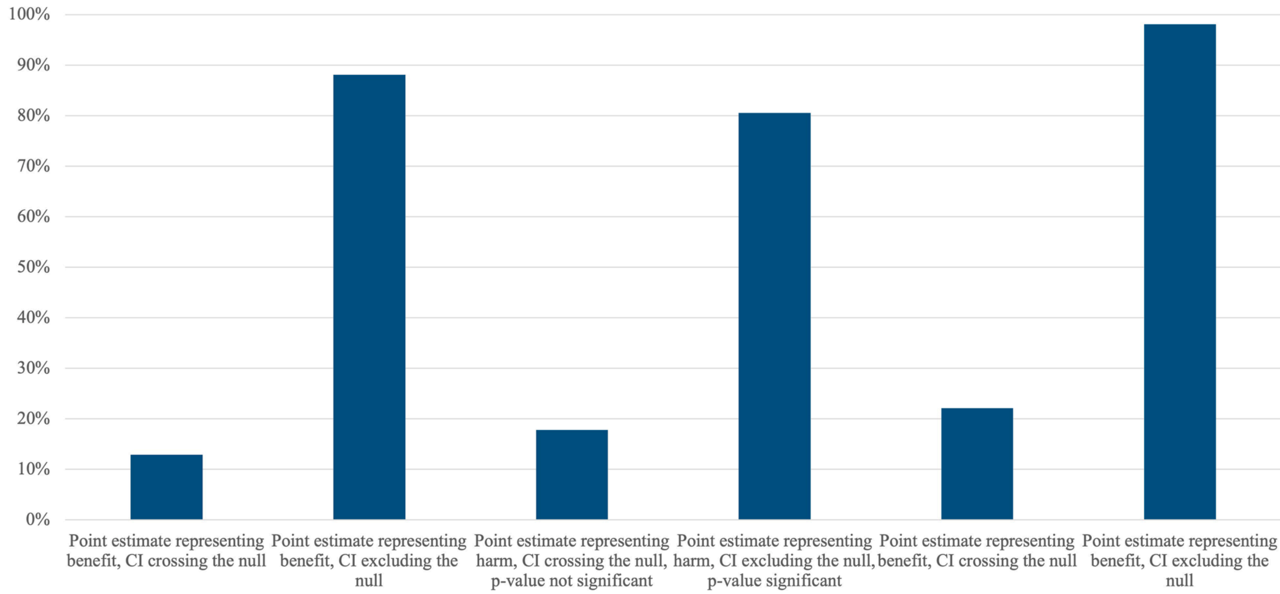

Figure 1. Proportion of participants with correct responses for cases with the null as the decision threshold and point estimate representing benefit or harm. CI, confidence interval.

found that most respondents defaulted to interpretations of ''no effect'' when the CI crossed the prespecified decision threshold (in particular, when it crossed the null), even when the point estimate indicated a benefit or harm. This pattern persisted regardless of the location of the point estimate relative to decision threshold (ie, the null or the MID), and was unaffected by professional status, research methods training and prior leadership in systematic reviews.

When point estimates were directly on the null, most clinicians correctly interpreted the effect as ''no effect.'' However, when point estimates were close to the null, most based their judgements on the CI rather than on the point estimate, judging ''no effect'' when the CI crossed the null and presence of an effect when the CI did not. Findings were similar with the MID as the decision threshold, with fewer respondents correctly interpreting an effect close to the null when the CI was narrow than when it was wide.

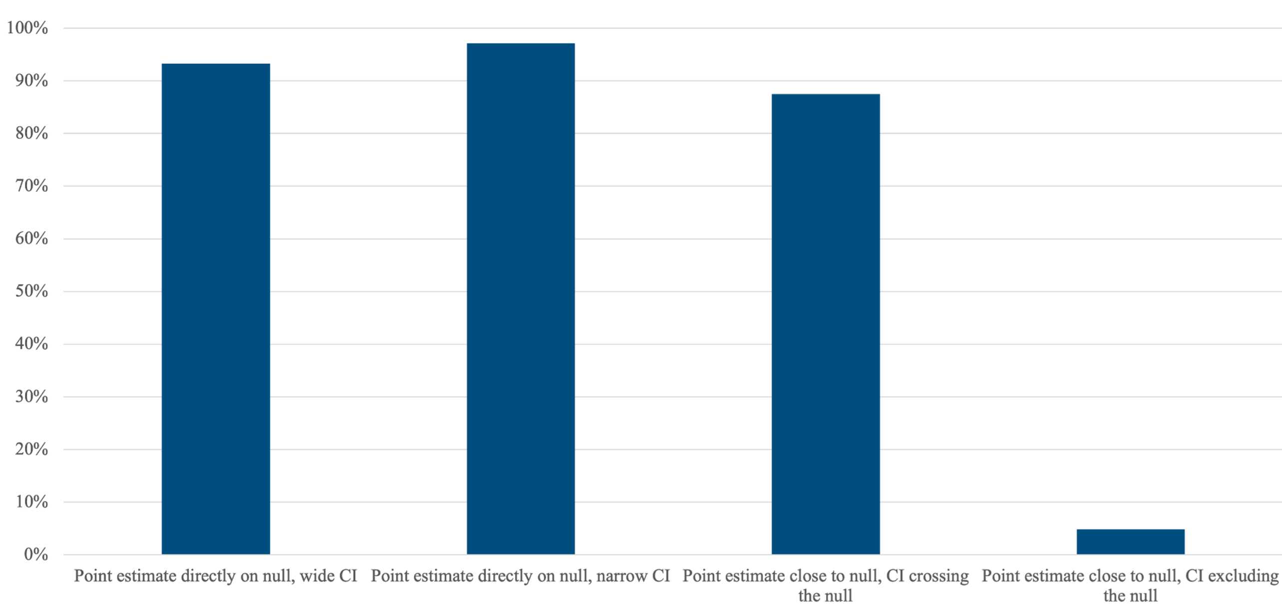

Figure 2. Proportion of participants with correct responses for cases with the null as the decision threshold and point estimate representing a null effect. CI, confidence interval.

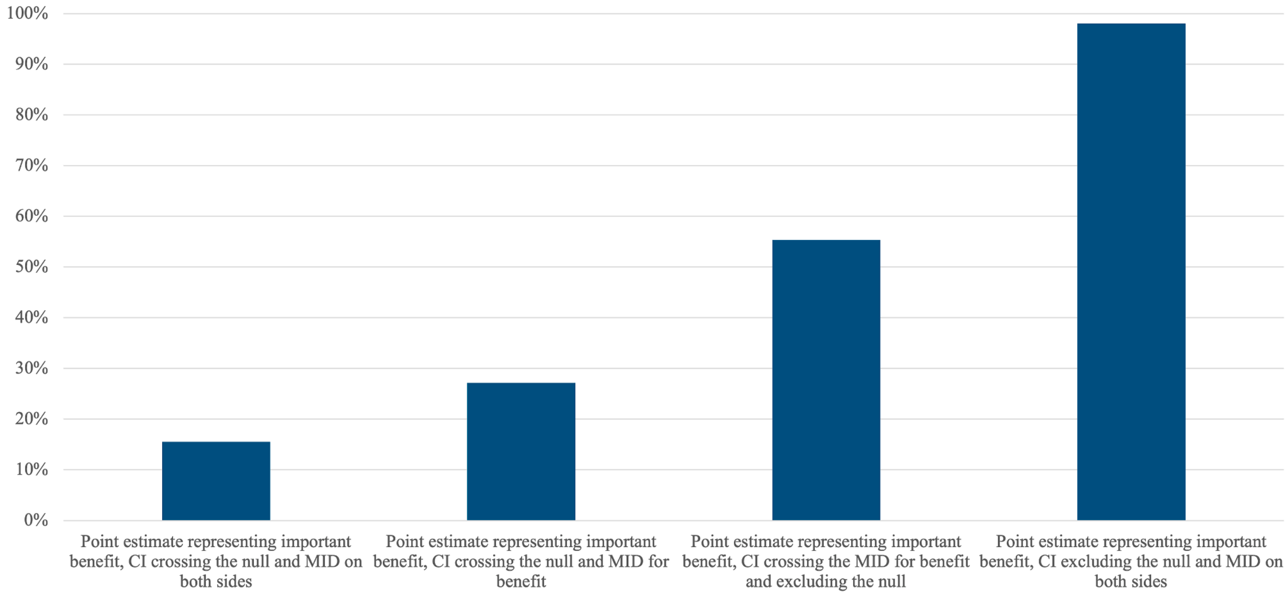

Figure 3. Proportion of participants with correct responses for cases with the MID as the decision threshold and point estimate representing an important benefit or harm. CI, confidence interval; MID, minimal important difference.

4.2. Comparison with existing evidence

These results align with prior work showing widespread misunderstanding of statistical significance and CIs among clinicians. Earlier surveys have documented poor comprehension of P values and effect sizes across multiple specialties [13–20], but no recent studies to our knowledge have directly isolated the specific error in equating CI overlap with the null to absence of effect. Our design—holding the point estimate constant while varying CI width and its relationship to the decision threshold—highlights the root of this misconception.

4.3. Interpretation and implications

Our findings are discouraging. In our survey, each case was accompanied by three response options (and an ‘‘other’’ option); by chance alone, each respondent would be expected to answer correctly 25% of the time, and 25% of respondents would be expected to answer correctly.

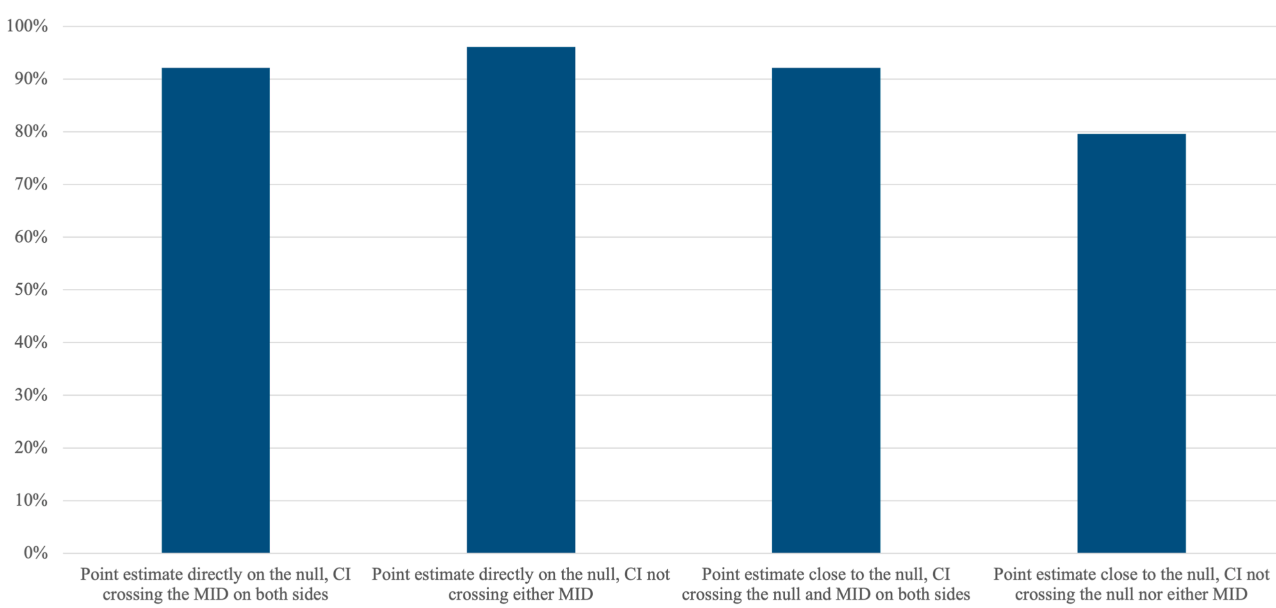

Figure 4. Proportion of participants with correct responses for cases with the MID as the decision threshold and point estimate representing an unimportant effect. CI, confidence interval; MID, minimal important difference.

However, less than 25% of respondents answered correctly for most cases, indicating that their responses were not random but reflected systematic misinterpretation of results that were not statistically significant.

Our findings are, however, not surprising and are in line with earlier studies. Despite significant coverage over the past decade cautioning against misconceptions regarding statistical significance and conflation of *P* values above a certain threshold or wide CIs that include the null indicating no effect or no difference [2,22,23], readers of the medical literature appear to persistently misinterpret such scenarios.

The GRADE approach, a systematic framework for rating the certainty of evidence, provides important guidance about interpreting a point estimate based on where it lies in relation to a prespecified decision threshold (whether the null threshold or MID), while using the CI to express the precision of that estimate—not to decide whether an effect exists. GRADE provides guidance on assigning a degree of (un)certainty in each summary effect estimate [9–12]. Beyond ensuring appropriate interpretation by clinicians, these findings have broader implications for guideline development, policy-making and patient communication. Clinical practice guidelines and shared decision-making tools should explicitly incorporate clear and accurate language when conveying treatment effects to patients [24–26].

4.4. Strengths and limitations

Our study has several strengths. The choice of treatment effect estimates presented covered scenarios where point estimates represented benefits harms and no effect. The survey encompassed both interpretation of effect estimates using the null and the MID as the decision threshold, both of which are commonly used in systematic reviews and meta-analyses and are likely to be encountered in real-world practice frequently. The survey included internal medicine residents and clinical fellows at varied levels of training, and faculty members across the Department of Medicine, enhancing generalizability.

Several important limitations exist. First, given the total number of individuals receiving the survey was unclear, we were unable to accurately calculate a response rate; a best estimate of our response rate was 8.8% with a total cohort of approximately 1240 individuals. Low response rates may result in a sample that inadequately represents all internal medicine trainees and academic faculty members. It is possible that respondents were more interested in research interpretation than non-respondents; if so, the true extent of misinterpretation among the broader group may be even greater. Second, our survey was limited to internal medicine trainees and faculty members at a single institution with a strong focus on evidence-based medicine (McMaster University), which may limit generalizability to institutions with different academic structures or lesser focus on research methods; one might expect greater misinterpretation at other institutions. Responses reflected greater representation from core internal medicine residents and full-time faculty members compared to from clinical fellows and part-time faculty members; while plausible this may limit generalizability to other specialties or to latter-mentioned groups, we do not have a strong basis to suspect results are likely to differ substantially. Third, while readers were instructed to make general interpretations and not focus on rating certainty of evidence, judgments related to how precise estimates were and the width of the CI may have influenced judgments. For instance, GRADE provides a framework for rating certainty in an effect estimate and suggests rating down certainty when the CI of a treatment effect crosses a chosen threshold. Respondents may have plausibly changed their responses based on concerns regarding how precise a presented effect estimate is. To guard against this, in addition to explicit instructions to not focus on rating certainty of evidence, we also included language in our response options that was compatible with varied levels of certainty in GRADE methods (for example, “benefit [or probably or possibly benefit],” where “probably” is consistent with moderate certainty, “possibly” with low certainty,” and the use of neither with high certainty). We therefore anticipate the effect of such judgments on our results would be minimal. Fourth, our survey did not evaluate how participants interpret effect estimates where a moderate or large effect threshold is used; we pragmatically limited the focus of our survey to the two decision thresholds we judged were most frequently used in systematic reviews and meta-analyses. Fifth, our multiple logistic regression analysis was likely underpowered due to a small sample size, sparse number of correct responses for some cases and insufficient variability in responses for others. Finally, real-world interpretation of treatment effects often incorporates baseline risk and absolute benefit. Our cases intentionally excluded these elements to isolate interpretation of point estimates and CIs relative to decision thresholds. Some respondents may have preferred a more nuanced judgment than the response options allowed; however, the consistent response patterns across cases suggest that misunderstanding of CIs, rather than clinical nuance, primarily explains the findings.

5. Conclusion

Clinicians—even those trained in evidence-based medicine—frequently misinterpret treatment effects, concluding “no effect” whenever a CI crosses the null. Educational efforts, journal policies and guideline processes should explicitly emphasize the interpretation of a point estimate based on its relation to a threshold and use the width of the CI to assess the certainty of the evidence. Our findings are limited to internal medicine trainees and faculty members at a single institution with a strong focus on evidence-based medicine.

CRediT authorship contribution statement

Arnav Agarwal: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Nancy Santesso:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Thomas Agoritsas:** Writing – review & editing, Methodology, Investigation. **Gordon Guyatt:** Writing – review & editing, Methodology, Investigation. **Romina Brignardello-Petersen:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

There are no competing interests.

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Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.jclinepi.2026.112206.

Data availability

All associated data is provided in the manuscript and accompanying supplementary files.

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